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TITLE OF THE EXPERIMENT :

To create a JavaScript-based student profile application using classes, objects, DOM manipulation, dynamic HTML elements, and event handling.

DESCRIPTION:

The Student Profile application is a simple web application developed using HTML, CSS, and JavaScript. It uses a JavaScript Student class with properties such as Name, Roll Number, Department, and CGPA. The user enters the student details through input fields, and when the Display Student Profile button is clicked, JavaScript collects the entered values using DOM methods. A Student object is then created using the class constructor. The program dynamically creates HTML elements using `createElement()`, adds student information using `textContent`, and displays the elements on the webpage using `appendChild()`. The `addEventListener()` method is used to handle the button-click event. Basic CSS is applied to make the dynamically generated student profile clear and attractive. Thus, the program demonstrates JavaScript classes and objects, DOM selection, dynamic HTML element creation, content modification, DOM manipulation, and event handling.

PROGRAM :

```
<!DOCTYPE html>
<html>
<head>
<title>Student Profile</title>
<style>
body {
font-family: Arial, sans-serif;
background-color: #f2f2f2;
text-align: center;
padding: 30px;
}
.container {
background-color: white;
width: 400px;
margin: auto;
padding: 25px;
border-radius: 10px;
box-shadow: 0 0 10px gray;
}
input {
```

```

width: 90%;
padding: 10px;
margin: 8px;
}
button {
padding: 10px 20px;
background-color: blue;
color: white;
border: none;
border-radius: 5px;
cursor: pointer;
}
#profile {
margin-top: 20px;
padding: 20px;
background-color: #e8f0fe;
border-radius: 10px;
text-align: left;
}
#profile h2 {
text-align: center;
}
</style>
</head>
<body>
<div class="container">
<h1>Student Profile</h1>
<input type="text" id="name" placeholder="Enter Name">
<input type="text" id="roll" placeholder="Enter Roll Number">
<input type="text" id="department" placeholder="Enter Department">
<input type="number" id="cgpa" step="0.01" placeholder="Enter CGPA">
<br>
<button id="displayBtn">Display Student Profile</button>
<div id="profile"></div>
</div>
<script>
// Creating Student class
class Student {
constructor(name, rollNumber, department, cgpa) {
this.name = name;
this.rollNumber = rollNumber;
this.department = department;
this.cgpa = cgpa;
}
}
// Select button using DOM
const button = document.getElementById("displayBtn");
// Event handling
button.addEventListener("click", function () {
// Get values from input fields
let name = document.getElementById("name").value;

```

```
let roll = document.getElementById("roll").value;
let department = document.getElementById("department").value;
let cgpa = document.getElementById("cgpa").value;
// Create Student object
let student = new Student(name, roll, department, cgpa);
// Select profile area
let profile = document.getElementById("profile");
// Clear previous content
profile.innerHTML = "";
// Create elements dynamically
let heading = document.createElement("h2");
heading.textContent = "Student Profile";
let nameElement = document.createElement("p");
nameElement.textContent = "Name : " + student.name;
let rollElement = document.createElement("p");
rollElement.textContent = "Roll No : " + student.rollNumber;
let departmentElement = document.createElement("p");
departmentElement.textContent = "Department : " + student.department;
let cgpaElement = document.createElement("p");
cgpaElement.textContent = "CGPA : " + student.cgpa;
// Add elements to webpage
profile.appendChild(heading);
profile.appendChild(nameElement);
profile.appendChild(rollElement);
profile.appendChild(departmentElement);
profile.appendChild(cgpaElement);
});
</script>
</body>
</html>
```

OUTPUT :

